

**The two main, indirect methods for finding exoplanets are the Doppler method and the transit method.**

The Doppler method or Radial Velocity method is based on the fact that the host star is pulled gravitationally by its orbiting planet, thus causing it to "wobble." As the star approaches us, its light is compressed into smaller wavelengths (blueshifted) and as it moves away, its light is spread into larger wavelengths (redshifted). This pattern can be used to measure the existence, mass and orbital period of a planet.

The transit method looks at the light emitted by a star over time. If the planet's orbital path occurs in between the host star and Earth, the planet will pass in front of the star thus obscuring a minute amount of the star's light. The periodic nature of these dips of the star's light can then determine the size of the planet, and observing the star's light through the planet's atmosphere at the time of a transit can allow for analysis of chemical composition.

**It is so difficult to see exoplanets directly in an image because of the overwhelming brightness of the host star compared to the planet.**

There are two fundamental difficulties in producing direct images; contrast and angular separation. Stars typically have brightness levels that are a billion times larger than that of a planet's reflected light at optical wavelengths. Attempting to image a planet situated beside a star is similar to photographing a small gnat around a massive searchlight from several miles away. The image will simply be overwhelmed by the glare from the star.

At the enormous distances of these star systems the planet will appear very close to its parent star from Earth. To detect such planets, it has been necessary to produce sophisticated tools such as coronagraphs to physically mask the star light or adaptive optics to remove the effects of the Earth's atmosphere. Even so it has currently been possible to directly image large young hot planets that orbit distant from their stars. Large because they produce their own thermal radiation and far distant because they orbit far from the glare to be observed separately.

**Similarities or differences between our Solar System and new, distant planet systems reveal that while some features are common, others are quite unexpected.**

One major disparity is that many planets like Hot Jupiters have been discovered. Our Solar System hosts large gaseous planets, such as Jupiter and Saturn, on the outskirts and beyond our sun. However, the initial few systems that were discovered seemed to contain gas giant planets orbiting very close to their respective star-even closer than Mercury to the sun! This realization did not concur with first planetary formation theories, and indicated that planets can migrate from where they form to much nearer orbits.

A key similarity on the other hand is the existence of terrestrial planets in the habitable zone. Many of these systems discovered, just like the Solar system, contain terrestrial planets at a distance from their star where liquid water would not freeze on their surface. These systems show us that the Solar system is perhaps not a very unusual phenomenon in that it possesses a rocky arrangement in the interior.